

Quantum Standard Basis

Description and Uses

1. Simple Definition

The **standard basis** (also called the *computational basis*) is the default set of reference states used to describe a qubit. For a single qubit it consists of two states, written in Dirac notation as $|0\rangle$ and $|1\rangle$. Every possible qubit state can be written as a weighted combination (superposition) of these two basis states.

As column vectors, the two basis states are:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These two vectors are **orthonormal**: each has length 1, and they are perpendicular to each other (their inner product is 0). That is exactly what makes them a valid basis.

KEY CONCEPT

$|0\rangle$ and $|1\rangle$ are the quantum analogue of the classical bits 0 and 1. The difference is that a qubit may exist in a superposition $\alpha|0\rangle + \beta|1\rangle$, whereas a classical bit must be strictly one or the other.

2. How and Why It Is Used

A general single-qubit state is written $\alpha|0\rangle + \beta|1\rangle$, where α and β are complex amplitudes satisfying $|\alpha|^2 + |\beta|^2 = 1$. The standard basis matters for several practical reasons:

Measurement reference	When you measure a qubit, the hardware reports the outcome in the standard basis. You get result 0 with probability $ \alpha ^2$ and result 1 with probability $ \beta ^2$.
A coordinate system	Just as x-y axes let you describe any point in a plane, $ 0\rangle$ and $ 1\rangle$ let you express any qubit state as a coordinate pair (α, β) .
Gate definitions	Quantum gates are written as matrices in this basis, so circuits are defined and simulated relative to it.
Multi-qubit indexing	For n qubits, basis states such as $ 00\rangle$, $ 01\rangle$, $ 10\rangle$, $ 11\rangle$ map directly onto binary numbers, which is why algorithms encode data this way.

3. Examples and Illustrations

Example A — Pure basis states

A qubit prepared as $|0\rangle$ always measures 0; one prepared as $|1\rangle$ always measures 1. No randomness.

Example B — Equal superposition

$$|+\rangle = (1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle$$
$$P(0) = |1/\sqrt{2}|^2 = 0.5 \quad P(1) = 0.5$$

This is what a Hadamard gate produces from $|0\rangle$: a 50/50 chance of each outcome when measured in the standard basis.

Example C — Two-qubit basis

Two qubits have four standard basis states, each a 4-entry vector:

State	Binary	Vector
$ 00\rangle$	0	[1, 0, 0, 0]
$ 01\rangle$	1	[0, 1, 0, 0]
$ 10\rangle$	2	[0, 0, 1, 0]
$ 11\rangle$	3	[0, 0, 0, 1]

Example D — A different basis, for contrast

The standard basis is not the only choice. The **Hadamard basis** uses $|+\rangle$ and $|-\rangle$ instead. Measuring the superposition from Example B in the standard basis gives random results, but in the Hadamard basis it gives a definite answer. The basis you pick determines what ‘definite’ means — which is why the standard basis is stated explicitly as the default reference.